Name: Solutions

Year 10 Science: Chemical Sciences Topic Test

HOLY CROSS COLLEGE

Date:		**Weighting:**	10%
Total Marks:	/44	**Percentage:**	%

Section A - Multiple Choice questions. (12 marks)

Please use the answer sheet to mark off the choices you have made.

1. Which of the following particles are used in determining the position of an element within the periodic table?
    a. protons only
    b. neutrons only
    c. electrons only
    d. protons and electrons

2. Sodium has an atomic number of 11. When in its ground state (i.e. it has a charge of zero and **is not** an ion) its electronic configuration is:
    a. 11
    b. 2,8,1
    c. 2,9
    d. 10,1

3. An atom has an electron configuration of 2,8,8,1. What group would it be in?
    a. 1 (I)
    b. 4 (IV)
    c. 18 (VIII)
    d. 5 (V)

MULTIPLE CHOICE ANSWERS	
1	A
2	B
3	A
4	B
5	D
6	C
7	B
8	D
9	C
10	A
11	A
12	B

4. Ionic bonding occurs between:
    a. metal atoms and other metal atoms
    b. metal and non-metal atoms
    c. non-metal atoms and other non-metal atoms
    d. lattices

5. Sulfur has an electron configuration 2,8,6. Use this information to determine the most likely charge of sulfide ions.
    a. +6
    b. -6
    c. +2
    d. -2

6. Which of the following statements is NOT true regarding noble gases?
    a. They are also known as inert gases.
    b. They are in group 18.
    c. They are very reactive
    d. They are very stable, tend not to react or bond with other atoms.

7. Which *product* in the following chemical reaction is dissolved in solution?

$$CaCO_3\,(s) + 2HCl\,(aq) \rightarrow CaCl_2\,(aq) + CO_2\,(g) + H_2O\,(l)$$

    a. HCl
    b. $CaCl_2$
    c. CO_2
    d. H_2O

8. Which of the following equations is balanced?
    a. $C_2H_6 + O_2 \rightarrow CO_2 + H_2O$
    b. $C_2H_6 + O_2 \rightarrow 2CO_2 + 3H_2O$
    c. $C_2H_6 + 7O_2 \rightarrow 2CO_2 + 3H_2O$
    d. $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$

9. When solutions of sodium chloride ($NaCl$) and silver nitrate ($AgNO_3$) are mixed, solid silver chloride ($AgCl$) forms. Which ions remain dissolved in the solution?
    a. Na^+ and Cl
    b. Ag^+ and Cl
    c. Na^+ and NO_3^-
    d. Ag^+ and NO_3^-

10. What effect would crushing solid reactants have on the rate of reaction?
    a. increase
    b. decrease
    c. no change
    d. not enough information

11. Which of the following is **not** a common property of ionic compounds (e.g. salt)?
    a. Malleability

b. Brittleness
c. Hardness
d. Conductive as liquids or when dissolved in water

12. The isotope 50-Vanadium has a mass number of 50. How many neutrons are present in this isotope?
    a. 23
    b. 27
    c. 50
    d. 73

Section B - Short Answer questions (32 marks).

Please write answers into the spaces provided.

1. Magnesium has an atomic number of 12. (5 marks)
   a. State how many protons would be in the nuclei of its atoms.

      12

   b. State how many electrons would be in each of its atoms (assuming the atom is neutral).

      12

   c. State the electron configuration of its atoms when in their ground state.

      2, 8, 2

   d. Predict the most likely charge of magnesium ions.

      2+

   e. The symbol for a sodium ion is Na^+. Use this format to state the symbol for magnesium ion.

      Mg^{2+}

      [1 mark each]

2. State the period and group number of the following elements: (4 marks)
   a. Calcium

      period 4 group II (2)

   b. Lithium

      period 2 group I (1)

   c. Fluorine

      period 2 group VII (17)

   d. Argon

      period 3 group VIII (18)

      [$\frac{1}{2}$ mark each]

3. Predict the most likely charges of any ions formed by the following elements. (4 marks)
   a. Calcium

      $2+$

   b. Lithium

      $1+$

   c. Fluorine

      $1-$

   d. Argon

      0 [1 mark each]

4. Determine the balanced formula equation for each of the reactions described below: (3 marks)
   a. $\underline{1}\ P_4 + \underline{5}\ O_2 \rightarrow \underline{2}\ P_2O_5$

   b. $\underline{2}\ KClO_3 \rightarrow \underline{2}\ KCl + \underline{3}\ O_2$

5. Write a balanced formula equation for each of the reactions described below. (4 marks)
   a. When aqueous solutions of lead(II) nitrate, $Pb(NO_3)_2$, and potassium iodide, KI, are mixed, the yellow solid lead(II) iodide, PbI_2, precipitates from the solution, while the soluble salt, potassium nitrate, KNO_3, remains in solution.

$$Pb(NO_3)_{2\,(aq)} + 2\,KI_{(aq)} \rightarrow PbI_{2\,(s)} + 2\,KNO_{3\,(aq)}$$

   b. An aqueous solution of ethanol, C_2H_5OH, is produced when a glucose, $C_6H_{12}O_6$, solution reacts in the presence of a yeast catalyst. Gaseous carbon dioxide, CO_2, is also produced.

$$C_6H_{12}O_{6\,(aq)} \rightarrow 2\,CO_{2\,(g)} + 2\,C_2H_5OH_{(aq)}$$

[Correct formulae - 1 mark
Balancing - 1 mark]

6. List four ways in which the rate of a reaction may be increased (4 marks).

a. Increase temperature

b. Increase concentration

c. Increase surface area

d. Use a catalyst

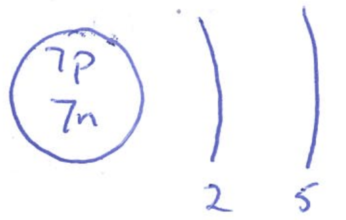
[1 mark each]

7. The solutions listed below are mixed. For each, state the precipitate formed. If not precipitate is formed write "N/A". (4 marks)

NaCl and $AgNO_3$	AgCl
CaI_2 and $PbNO_3$	PbI_2
K_2CO_3 and $CaCl_2$	$CaCO_3$
$CuSO_4$ and $FeCl_2$	NA

8. Draw diagrams to show the electron configurations of the following elements. **Remember to include the particles in the nucleus.** (4 marks)

a. $^{14}_{7}N$



[Nucleus - 1 mark
Electrons - 1 mark]

b. $^{35}_{17}Cl^{-}$